

Virtual Rolling Temple: Expanding the Vertical Input Space of a Smart Glasses Touchpad

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ABSTRACT

Smart glasses have not favored two-dimensional (2D) GUI. Such a trend may have originated from the limitations of smart glasses in display and input devices. While the display restriction is rapidly being resolved nowadays, 1D GUI is still the majority, indicating that the touch input device is the possible bottleneck. To tackle this issue by expanding the vertical input space of the temple touchpad, we propose the Virtual Rolling Temple (VRT). The concept is to perform 2D gestures by moving the hand in any direction while keep touching the prototype as if the temple rotates. The VRT touchpad is as thin as the spectacles' temples, but it provides the users with input space approximately equivalent to an 80×80 mm square touchpad. This is 8 and 13.9 times larger than Google Glass and VUZIX M400, respectively. To validate the concept of the VRT, we constructed three demo scenarios: 2D Pointing, 2D Menu, and 2D Gesture, to cover different types of general 2D input for smart glasses.

CCS CONCEPTS

• Human-centered computing → Pointing devices; Mixed / augmented reality.

KEYWORDS

Smart glasses, pointing device, two-dimensional input

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1 INTRODUCTION

Two-dimensional (2D) GUI is common in major computing platforms, such as desktops and smartphones. On the other hand, the GUI of smart glasses is often one-dimensional (1D), organized linearly along the horizontal axis. We suppose this difference may be due to the limitations of smart glasses in their display and input device.

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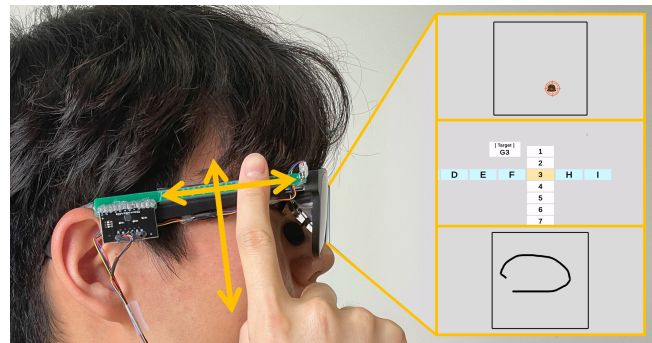


Figure 1: The Virtual Rolling Temple (VRT) attached to NREAL Air AR glasses. The picture illustrates how the user can move the finger on a narrow temple for 2D input.

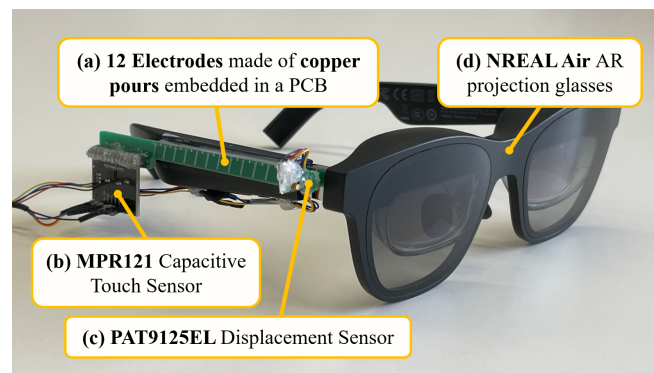


Figure 2: Prototype with labels on its major components.

First, it may be because of smart glasses display's narrow field of view (FoV). For example, Google Glass (2013) shows one page at a time, such as a clock, weather, and compass, through the 14.8° FoV display [1, 3]. The VUZIX M400 (2020) has 16.8° FoV [8]. It can show more items on the screen, but its menu interface is still 1D [6]. However, the FoV of smart glasses is increasing rapidly and may no longer be the reason for favoring 1D GUI. For example, among glasses-shaped devices, VUZIX Shield (2022) has 30° FoV, and NREAL Air (2022) achieved 46° FoV [4, 7].

Another reason for 1D GUI may have been the narrow form factor of the touchpad on the temple. It could have restricted vertical input and acted as a pressure to favor 1D GUI. For instance, Google Glass has an 80×10 mm touchpad on the temple, and its Timeline is a horizontal arrangement of multiple rectangular pages [1]. It

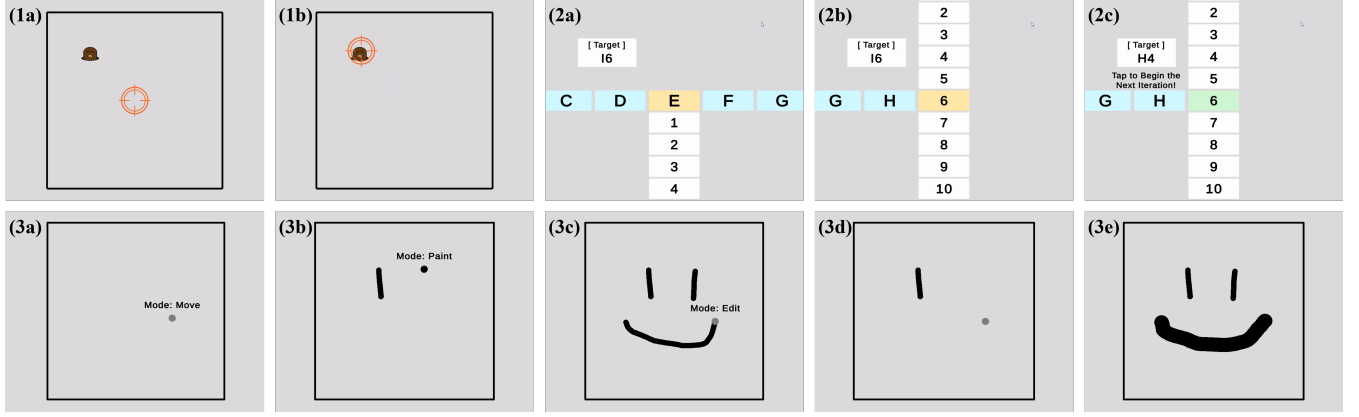


Figure 3: Visualization of the three demonstration scenarios mentioned on the previous page. (1) 2D Pointing, (2) 2D Menu, and (3) 2D Gesture. Each scenario is chosen to cover different types of general 2D input for smart glasses.

does not use vertical gestures except for a few cases, such as closing the pop-up window. VUZIX M400 is equipped with a 33×14 mm touchpad. It supports both horizontal and vertical gestures [9], but the control-display (c-d) gains for the two axes differ significantly to compensate for the limited vertical input space. Researchers also mentioned that the thin form factor makes it difficult to distinguish among the eight directional gestures [2].

To expand the vertical input space of the temple touchpad and thereby better support 2D GUI on smart glasses, we propose the Virtual Rolling Temple (VRT). The user can perform 2D inputs by moving the hand in any direction while keeping a finger or fingers on the temple as if he/she rotates the temple, as shown in Figure 1. The user can vertically move the hand up to the finger’s length without clutching. It means that the vertical input space of the temple touchpad is expanded to the finger’s length.

2 VIRTUAL ROLLING TEMPLE (VRT)

The VRT may be implemented by adding a physical roller on the temple. However, a physical roller on a temple may pose many technical problems. In addition, it may hinder the smart glasses designs from being aesthetic and monolithic. Therefore, we “virtually” implemented the VRT utilizing touch and displacement sensors.

We designed a printed circuit board (PCB) with 12 touch-sensing electrodes in an 80×6 mm touchpad area. We hoped to minimize the PCB touchpad height to mimic real temples but chose the 6 mm considering our prototyping skill. The PCB was then attached to the right temple of NREAL Air AR glasses (Figure 2(d)).

The 12 electrodes on the PCB were for sensing horizontal position (Figure 2(a)). The electrodes were connected to the MPR121 12-channel capacitive touch sensor (Figure 2(b)), which was directly soldered to the PCB to minimize parasitic capacitance. We then interpolated the 12-channel sensor values to estimate the horizontal position of the finger. To reduce friction in the touch area, we covered it with Scotch Magic Tape.

To detect the movement along the vertical axis, we used a laser speckle displacement sensor (PAT9125EL, Figure 2(c)). To increase its operating distance, we disabled the internal laser and added an external one (OPV322). With this modification, the sensor could

stably operate at up to 200 mm distance, sufficiently covering the width of the touchpad area. The sensor was placed at the frontal end of the PCB, facing backward, to detect the vertical finger movement. The resulting resolution was 26 counts/cm. The vertical finger movement sensing was enabled only when the finger was on the touchpad area.

We estimated the hand motion on a virtual 2D plane by combining the readings of the two sensors. On average, the sensing and transmission phase achieved a frame rate of around 150 frames/s.

3 DEMO SCENARIOS

We propose three demo scenarios: 2D Pointing, 2D Menu, and 2D Gesture, as shown in Figure 3. In all scenarios, the mapping from the VRT and the GUI was as follows. The user swipes forward, backward, up, or down, to move the cursor right, left, up, and down, respectively. The c-d gains for the two axes are the same; therefore, the input space of the VRT is equivalent to an 80×80 mm square touchpad.

3.1 2D Pointing: Catch a Mole

The user can navigate the cursor to the mole and tap to catch it (Figure 3(1b)). The mole appears on a plane of 20 by 20 meters, placed 20 meters away, facing the user. When it is caught, it moves to a random position, and the cursor is restored to the center (Figure 3(1a)). This design was to mimic the Fitts’ Law test [5].

3.2 2D Menu: Two-Level Menu

The user can scroll horizontally to navigate through nine categories and vertically to select among ten items in each category (Figure 3(2a)). At the chosen c-d gain, the user can navigate to any category or menu item at once with a single gesture consisting of subsequent horizontal and vertical strokes. In order to minimize searching, categories are labeled with letters from A to I, and menu items are labeled with integers from 1 to 10. In each trial, a target is randomly determined and displayed in the text box within the FoV. The user then reads the target and taps to start the trial. The user scrolls toward the target (Figure 3(2b)) and taps to select it (Figure 3(2c)).

3.3 2D Gesture: Paint

The user can draw lines in any shape. The demo has three modes: Move, Paint, and Edit, where the brush is visualized as black in the Paint mode and gray in the other modes. The user can cycle through the three modes by performing a two-finger tap. When the mode is switched, the new mode name is displayed above the brush for a short period (Figure 3(3a, 3b, 3c)).

One-finger gestures move the brush in the Move mode (Figure 3(3a)) and draw along the trace of the brush in the Paint mode (Figure 3(3b)). In the Edit mode, the user can undo or redo the drawing, stroke by stroke, by moving forward or backward on the VRT (Figure 3(3c, 3d)). The user can also move vertically on the VRT to change the brush size within a preset range (Figure 3(3e)).

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